学　号
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西北工业大学研究生学位论文选题报告表

Northwestern Polytechnical University
Graduate Degree Dissertation Title Selection Report Form

学　院	计算机学院
School	School of Computer Science
学科、专业	计算机科学与技术
Discipline / Specialty	Computer Science and Technology
姓　名	**DAULET ANEKESH** (大雷)
Name	DAULET ANEKESH
学 位 级 别	硕士
Degree	Master
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培 养 类 别	留学研究生
Category	Overseas Postgraduates
报 告 日 期	2026年10月
Date of Report	October 2026

研　究　生　院

Graduate School

研究生学位论文选题报告的要求

一、 硕士生的选题报告内容应包括文献综述、选题意义、研究内容、研究方案，论文工作量的估计、工作条件

二、 博士生的选题报告内容应包括文献综述、选题背景及其意义、研究内容、研究特色、工作难点、预期成果

三、 选题报告会应以学术活动的方式公开进行。

四、 正式开题之前，研究生应在广泛阅读中、外文资料的基础上，深入了解拟选课题的国内外研究动态，把握

五、 研究生在规定的时间内，写出选题报告初稿，经指导教师审阅同意后，由指导教师安排选题报告时间。选

六、 选题报告不能按期完成者，应及时向研究生院培养处提出延期申请。

七、 本表可以打印或用钢笔认真填写，若不够填写时，可另加附页。

Requirements for the Dissertation Title Selection Report

1. The Dissertation Title Selection Report for Master Degrees should include the Literature Review, the Meaning of the Title Selection, the Research Contents, the Research Plan, Workload Estimation, Working Conditions, the Goal to Achieve, Problems Existing as well as the Measures to Take.
2. The Dissertation Title Selection Report for PhD Degrees should include Literature Review, Background and Meaning of the Selection, Research Contents, Characteristics of the Research, Difficulties, Goals to Achieve and possible Innovations.
3. The Report should be made in public as academic activities.
4. Before starting the title, graduates should, based on the extensive reading through the literature both in Chinese and in foreign languages, have a deep command of the research tendency home and abroad. They must have a clear mind of the aim, meaning and foreseen results, research assumption, methods and approaches.
5. Graduates should finish the draft report within the required time, and then the supervisor after having checked the report, organizes time to make the report. Those who fail to pass the report have to reopen a title. If it can not be passed for a second time, the studentship will be stopped.
6. Those who can not finish the report on time should forward application to the Graduate School to extend the duration.
7. The form should be filled out in pen or printed. If the pages are not enough, extra pages can be attached.

论文题目 **Dissertation Title**	**Morphologically-Grounded AI-Generated Text Detection and Factual Verification for Low-Resource Agglutinative Languages: A Comprehensive Framework for Kazakh**			
论文类型 (在有关项目下作记号√) **Dissertation Type** (Put a tick √ under relevant item)	基础研究 Fundamental Research	应用研究 Application Research √	工程技术 Engineering Technology	跨学科研究 Research Cross Discipline
研究生在进行选题报告、听取意见后，整理成文（不少于 3500 字）。 **After the report and comments, make the records up to an article. (no less than 3500 words)**				

1 Research Background and Significance

The recent proliferation of autoregressive Large Language Models (LLMs)—such as OpenAI's GPT-4, Meta's LLaMA-3, and DeepSeek-R1—has precipitated a fundamental paradigm shift across artificial intelligence and digital communication [1–4]. While these models exhibit unprecedented proficiency in syntactic cohesion, creative writing, and automated reasoning, their widespread accessibility has simultaneously intensified severe risks surrounding information integrity [5]. Modern digital ecosystems are increasingly inundated with synthetic media, including automated disinformation campaigns, AI-assisted academic plagiarism, and hallucinated factual claims that masquerade as authoritative knowledge. Addressing these existential challenges demands robust computational mechanisms for both (1) distinguishing machine-generated prose from human-authored text (AI text detection), and (2) verifying the empirical veracity of factual assertions against reputable reference knowledge (automated fact verification).

Despite vigorous progress in mainstream languages such as English and Standard Mandarin, modern text safety and verification methodologies encounter critical operational bottlenecks when deployed in low-resource, morphologically complex agglutinative languages [6]. Languages belonging to the Turkic family—exemplified by Kazakh, Uzbek, Kyrgyz, and Turkish—exhibit structural typologies that fundamentally contradict the core architectural assumptions of prevailing transformer models. In agglutinative linguistic systems, lexical roots serve as anchors for extensive, sequential suffixation chains that systematically encode negation, evidentiality, syntactic case relations, tense, possessive markers, and pragmatic moods within single polymorphemic words. A solitary verbal stem can generate hundreds of distinct surface forms without introducing new lexical headwords.

This structural reality produces three catastrophic vulnerabilities in standard natural language processing (NLP) pipelines:

(1) **Extreme Lexical Sparsity and Token Fragmentation:** Standard Byte-Pair Encoding (BPE) and WordPiece tokenizers, trained predominantly on high-resource multilingual corpora, lack structural awareness of Turkic morphological boundaries. Subword tokenizers aggressively fragment agglutinative suffix chains into meaningless sub-character pieces (e.g., segmenting the Kazakh word *мектебіміздегілерден* ["from those who are in our school"] into multiple disjoint byte tokens). This fragmentation artificially inflates sequence lengths, depletes effective context windows, and dilutes gradient flow during representation learning.

(2) **The Failure of Surface Lexical Overlap Heuristics:** Conventional sparse information retrieval engines (e.g., BM25) and sentence similarity baselines operate on exact surface token matches [7].

Because Kazakh inflections drastically alter surface tokens while preserving core semantic lemmas, raw keyword matching suffers from severe recall degradation. Conversely, in natural language inference (NLI), minor morphological shifts can completely invert semantic truth. For example, altering a single unstressed two-character negative infix (*келген* ["arrived"] vs. *келмеген* ["did not arrive"]) reverses factual polarity while maintaining an 80% surface lexical overlap, tricking standard classifiers into predicting factual entailment when the proposition is explicitly contradicted.

(3) **Decoupled Veracity and Provenance Silos:** In existing industrial and academic auditing frameworks, AI text detection and factual verification are treated as disjoint, independent tasks. In real-world media environments, however, information risks manifest across both axes simultaneously: human-written texts may contain malicious falsehoods (human misinformation), AI models can synthesize completely fabricated assertions (AI disinformation), while AI systems may also summarize factual ground truths (AI synthesis). Decoupled systems fail to provide human fact-checkers with a unified, actionable assessment of content reliability.

To bridge these critical gaps, this research establishes the first unified, morphologically-grounded research framework tailored specifically for low-resource agglutinative languages, taking Kazakh as the primary exemplar. By integrating formal computational morphology (an 83-rule Finite-State Transducer) into deep contrastive detection and hybrid retrieval-augmented verification, this dissertation bridges linguistic theory and modern deep learning. Furthermore, this research introduces the first human-validated benchmark for Kazakh fact verification (**Kazakh-FEVER 3K**) equipped with a specialized "Hard Not-Enough-Info" (Hard NEI) protocol, and formulates a continuous 2D Cartesian Trust Matrix $(T_{\text{fact}}, T_{\text{gen}})$ that unifies factual truth and generative provenance into an interpretable decision space. This study provides vital foundational infrastructure for linguistic equity, sovereign information verification, and trustworthy AI deployment across Central Asia and the broader Turkic-speaking world.

2 Literature Review and Research Status

The intersection of artificial intelligence detection, factual verification, and computational morphology represents an active, rapidly evolving frontier in modern natural language processing. This section surveys current international and domestic research across these five foundational domains.

2.1 AI-Generated Text Detection Methodologies

Automated detection of machine-generated text has advanced through three successive methodological waves: metric-based statistical detectors, supervised neural classifiers, and perturbation-based curvature analyzers.

Early statistical techniques leveraged intrinsic token distribution properties, evaluating n-gram perplexity, rank order frequencies, and burstiness under assumed language model priors [8]. These metrics capitalize on the empirical observation that autoregressive generation strategies (e.g., top-p nucleus sampling, top-k truncation) disproportionately select high-probability tokens from truncated distributions, yielding uniformly low perplexity trajectories compared to human text. Supervised neural classifiers subsequent to RoBERTa and DeBERTa approached detection as a standard binary sequence classification problem, learning discriminative stylistic artifacts directly from labeled datasets [9, 10]. However, supervised baselines suffer from severe distribution shifts: classifiers trained on older generator architectures (e.g., GPT-2) exhibit catastrophic performance degradation when tested on newer, instruction-tuned foundation models (e.g., LLaMA-3, Claude, GPT-4).

To eliminate the dependency on static training distributions, recent efforts have focused on zero-shot, training-free detection paradigms. Mitchell et al. [11] introduced DetectGPT, positing that machine-generated texts occupy local optima within the model's log-probability curvature landscape. By applying mask-filling perturbations (via T5) and measuring the directional log-likelihood divergence between perturbed and original sequences, DetectGPT achieves competitive zero-shot detection. Hans et al. [12] developed Binoculars, which formulates detection as the normalized ratio of cross-perplexity between two distinct pre-trained foundation models, removing the latency of stochastic sampling. Meanwhile, Bao et al. [13] presented Fast-DetectGPT, utilizing analytical approximations of the Hessian curvature to accelerate inference speeds by three orders of magnitude.

Despite their algorithmic elegance, virtually all contemporary detection benchmarks (e.g., HC3, M4, Ghostbuster [14], RADAR [15]) operate almost exclusively on English, German, or Mandarin Chinese. In low-resource agglutinative languages such as Kazakh, zero-shot curvature scoring fails because the surrogate scoring models (such as multilingual GPT-2 or mBERT) possess inadequate parametric vocabularies for Turkic languages, producing noisy, unstable log-probability surfaces that destroy the detection signal.

2.2 Automated Fact Verification and Retrieval-Augmented Pipelines

Automated fact verification emerged as a structured NLP discipline following the seminal release of the Fact Extraction and VERification (FEVER) benchmark by Thorne et al. [16]. The standard FEVER pipeline decomposes claim verification into a three-stage architecture: (1) *Document Retrieval*, identifying relevant Wikipedia articles from a large knowledge repository; (2) *Sentence Evidence Extraction*, selecting the minimal set of sentences sufficient to substantiate or refute the claim; and (3) *Claim Verification / Natural Language Inference*, predicting a three-way veracity label: Supported, Refuted, or Not Enough Info (NEI).

Subsequent benchmarking expanded the FEVER paradigm into specialized domains and real-world multi-source contexts. FEVER 2.0 [17] introduced adversarial human attacks; MultiFC [18] and Snopes-based datasets gathered claims from public fact-checking organizations; and CLIMATE-FEVER [19] explored climate change assertions characterized by subtle scientific nuances and conflicting evidence. In the retrieval tier, early systems relied on sparse inverted index mechanisms, notably BM25 [7]. The introduction of Dense Passage Retrieval (DPR) [20] and unsupervised contrastive retrievers (Contriever [21]) established dense neural embeddings as the dominant standard for semantic matching.

However, non-English fact verification benchmarks remain exceptionally sparse. While multilingual extensions such as X-FEVER, Multilingual FEVER, and FactCheck-Bench [22] incorporate translation-based evaluations, they overwhelmingly rely on synthetic machine translation of English benchmarks. These translated datasets systematically obscure indigenous cultural phenomena, local legal nuances, and native syntactic structures. Most critically, until this dissertation, there has been no dedicated, human-annotated fact verification benchmark for Kazakh or any other Central Asian language.

2.3 Computational Morphology for Agglutinative Languages

Computational modeling of agglutinative morphology has historically relied on two distinct traditions: formal symbolic rule engines and statistical/neural subword segmenters.

In the symbolic paradigm, Finite-State Transducers (FSTs) based on the two-level morphology framework (Koskenniemi) provide exact, bidirectional mappings between surface wordforms and grammatical analyses [23, 24]. The Apertium collaborative platform developed open-source morphological analyzers for Kazakh, utilizing extensive dictionaries and morphotactic continuation lexicons to handle vowel

harmony (palatal/labial harmony) and consonant assimilation [23]. While symbolic FSTs achieve near-perfect precision on in-vocabulary lexical stems, their integration with modern deep learning representations has been severely constrained by non-differentiable execution boundaries and brittleness when encountering out-of-vocabulary named entities or internet slang.

In the statistical paradigm, unsupervised subword algorithms such as Byte-Pair Encoding (BPE), WordPiece, and SentencePiece dominate modern language modeling. Although computationally efficient, recent studies have demonstrated that subwords introduce severe cognitive biases in agglutinative contexts. Khurana et al. [6] demonstrated that subword tokenizers frequently partition grammatical suffixes in Turkic and Uralic languages across arbitrary phoneme boundaries, completely obscuring syntactic dependencies and negation allomorphs. Hybrid frameworks that explicitly inject symbolic morphological features into neural embedding layers represent an urgent, under-explored research trajectory.

2.4 Shortcut Learning and the "Hard NEI" Pathology

A pervasive failure mode in contemporary natural language inference and fact-checking models is their vulnerability to *shortcut learning* (also known as spurious correlation exploitation) [22]. In standard FEVER benchmarks, instances belonging to the NOT ENOUGH INFO (NEI) class are typically created by pairing claims with unrelated passages retrieved via random sampling. Consequently, neural verifiers rapidly discover a superficial statistical shortcut: if a claim and an evidence passage exhibit low surface token overlap or distant semantic embeddings, the pair can be reliably classified as NEI with high training accuracy.

In real-world media environments, however, false or unverified claims are rarely disjoint from authentic facts. Malicious disinformation and subtle rumors routinely retain high lexical overlap with genuine news reports, altering only temporal anchors, quantitative bounds, evidential modalities, or logical scopes. When evaluated on such challenging instances, standard cross-encoders suffer catastrophic performance collapse, predicting SUPPORTED merely because the candidate evidence shares extensive topical vocabulary with the claim. Neutralizing this pathology requires explicit counterfactual curation protocols that demand fine-grained morphological reasoning.

2.5 Explainable AI and Multidimensional Trust Frameworks

In high-stakes domains such as investigative journalism, intelligence auditing, and content moderation, automated black-box predictions are fundamentally insufficient. Human auditors require transparent, interpretable rationales and calibrated uncertainty metrics before accepting model recommendations.

Feature attribution methods such as Integrated Gradients, Grad-CAM [25], and SHAP [26] provide post-hoc saliency heatmaps highlighting influential tokens. However, in low-resource agglutinative languages, token-level heatmaps are notoriously difficult to interpret when subwords segment suffixes into non-intuitive byte fragments. Furthermore, existing NLP benchmarks evaluate veracity (true vs. false) and generation origin (human vs. AI) in complete isolation. As generative AI becomes widely adopted by legitimate authors and malicious actors alike, automated decision systems require a unified, multi-criteria trust space that simultaneously captures factual accuracy and generative provenance, establishing an interpretable foundation for human-in-the-loop verification [27].

3 Research Objectives and Key Scientific Questions

3.1 Primary Research Objective

The overarching objective of this dissertation is to design, implement, and empirically validate an end-to-end, morphologically-grounded computational framework for artificial intelligence-generated text de-

tection and automated factual verification tailored specifically for low-resource agglutinative languages, using Kazakh as the primary benchmark language.

By unifying symbolic morphological transducers, contrastive representation learning, hybrid lexical-neural retrieval, and a novel 2D Cartesian trust formulation, this research aims to neutralize shortcut learning pathologies, overcome subword tokenization bottlenecks, and deliver an interpretable, production-ready auditing system for low-resource NLP.

3.2 Specific Objectives

To achieve the primary research goal, the project is structured around four interconnected specific objectives:

Obj 1: Symbolic Morphological Engineering and Feature Formulation: Design and construct an 83-rule Finite-State Transducer (FST) rule engine and stemmer for Kazakh that accurately isolates canonical root morphemes from complex inflectional and derivational suffixation chains, and formulate a 16-dimensional continuous affix diagnostic vector ($\mathbf{m}_{\text{affix}} \in \mathbb{R}^{16}$) capturing negation mismatches, temporal conflicts, evidential modalities, and syntactic case alignments.

Obj 2: Morpho-Contrastive AI Text Detection Pipeline: Develop a deep contrastive neural architecture that integrates pre-trained multilingual transformer backbones with morphological projection heads. Evaluate model invariance across a newly curated Tri-Domain corpus (formal news, encyclopedic text, and conversational social media) under strict Leave-One-Generator-Out (LOGO) cross-validation against diverse LLM families (LLaMA, GPT, Mistral, Sherkala-7B).

Obj 3: Construction and Validation of the Kazakh-FEVER 3K Benchmark: Curate the first human-validated, multi-domain fact verification benchmark for Kazakh comprising 3,000 claim-evidence pairs. Establish the novel *Hard Not-Enough-Info (Hard NEI)* curation protocol to enforce high lexical overlap ($\geq 85\%$) while varying truth-conditional morphological markers, and construct a hybrid evidence retrieval engine combining morpheme-stemmed BM25 with dense multilingual embeddings (mContriever) via Reciprocal Rank Fusion (RRF).

Obj 4: Mathematical Formulation of the 2D Cartesian Trust Matrix and Deployment: Formulate a unified continuous 2D trust metric $\mathcal{T}(x) \in [0, 1]$ that jointly projects factual veracity ($T_{\text{fact}} \in [-1, 1]$) and generative authenticity ($T_{\text{gen}} \in [0, 1]$) into four operational risk quadrants, and deploy a production-grade, bilingual (Kazakh and English) interactive web platform with real-time morphemic stem highlighting for human-in-the-loop auditing.

3.3 Core Research Questions

This investigation is guided and rigorously validated through four central research questions:

- **RQ1 (Morphological Representation):** *To what extent does the explicit injection of symbolic morphological diagnostic features ($\mathbf{m}_{affix}$) overcome the representation dilution caused by subword tokenizer fragmentation in agglutinative neural models?*
- **RQ2 (Retrieval Synergy):** *Can hybrid retrieval combining rule-based morphemic BM25 indexing with dense unsupervised representations (mContriever) achieve superior Recall@k and Mean Reciprocal Rank (MRR) across highly inflected low-resource queries compared to purely dense or purely sparse baselines?*

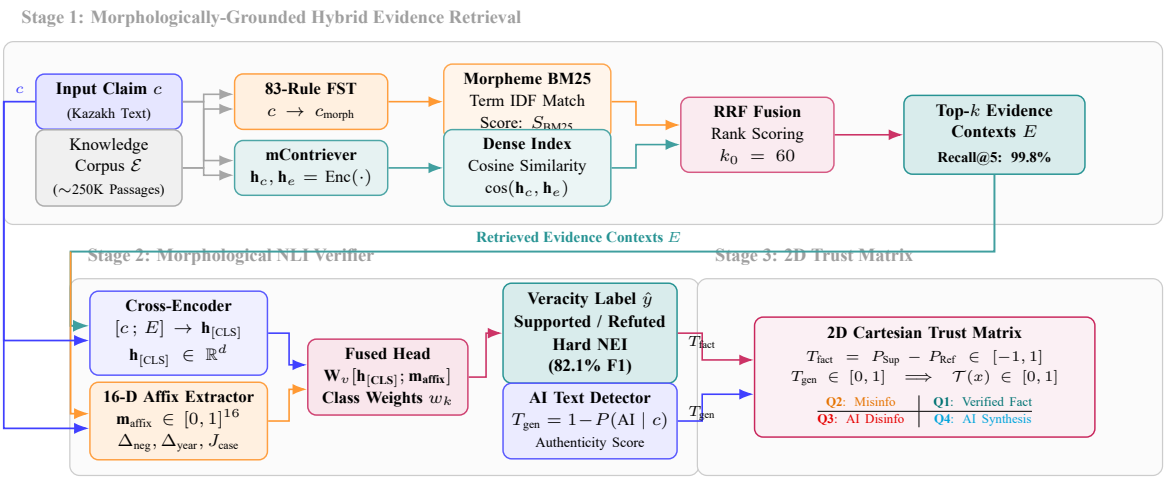


Figure 1: End-to-end system architecture for morphologically-grounded fact verification and AI text detection in low-resource agglutinative languages. **Stage 1**: 83-rule FST stemmer and dense multilingual encoder (MCONTRIEVER) fused via RRF ($k_0 = 60$) over a 250K passage knowledge corpus. **Stage 2**: Cross-encoder representation $\mathbf{h}_{[\text{CLS}]}$ fused with the 16-D morphological diagnostic vector $\mathbf{m}_{\text{affix}}$ via class-weighted softmax head (w_k). **Stage 3**: Continuous 2D Trust Matrix mapping factual veracity T_{fact} and generative authenticity T_{gen} into four operational risk quadrants.

- **RQ3 (Shortcut Neutralization):** *Does the Hard NEI curation protocol successfully eliminate superficial lexical overlap shortcuts in natural language inference classifiers, and does explicit morphological conditioning prevent false entailment predictions when truth-conditional allomorphs (e.g., verbal negation infixes) are manipulated?*

- **RQ4 (Trust Space Calibration):** *Does the joint 2D Cartesian Trust Matrix (T_{fact}, T_{gen}) provide well-calibrated, separable operational decision boundaries across verified human ground truth, human misinformation, AI synthesis, and AI disinformation compared to standard independent thresholding?*

4 Methodology and System Design

The methodological framework designed in this dissertation unites symbolic linguistic engineering with deep representation learning into an end-to-end, multi-stage processing pipeline. The complete system architecture is illustrated in Figure 1. The framework comprises four interconnected operational tiers: (1) Linguistic Preprocessing and the 83-Rule FST Morphological Engine, (2) Morpho-Contrastive AI Text Detection, (3) Morphologically-Grounded Fact Verification and Hybrid Retrieval, and (4) the Continuous 2D Cartesian Trust Matrix.

4.1 Linguistic Preprocessing and the 83-Rule FST Analyzer

Kazakh grammar adheres to strict phonological and morphotactic constraints, governed by palatal/labial vowel harmony and regressive/progressive consonant assimilation. To model these phenomena deterministically, we implement an 83-rule Finite-State Transducer (FST) rule engine that operates directly over normalized Cyrillic-Kazakh text.

Given an input string s, the transducer applies sequential peeling passes across four canonical inflectional tiers:

$$s \xrightarrow{\text{FST Pass 1}} s_{\text{case}} \xrightarrow{\text{FST Pass 2}} s_{\text{poss}} \xrightarrow{\text{FST Pass 3}} s_{\text{plur}} \xrightarrow{\text{FST Pass 4}} \mathcal{R}(s) \tag{1}$$

where $\mathcal{R}(s)$ denotes the canonical invariant root morpheme. The 83 grammatical rules encode:

- **Plural Suffixes (6 allomorphs):** *-лар/-лер*, *-дар/-дер*, *-тар/-тер*.
- **Possessive Suffixes (16 allomorphs):** 1st, 2nd (polite/informal), and 3rd person singular and plural markers (*-ым/-ім*, *-ыңыз/-іңіз*, *-сы/-сі*).
- **Case Suffixes (14 allomorphs):** Nominative (unmarked), Accusative (*-ны/-ні/-ды/-ді/-ты/-ті*), Dative (*-ға/-ге/-қа/-ке*), Locative (*-да/-де/-та/-те*), Ablative (*-дан/-ден/-тан/-тен*), Instrumental (*-мен/-бен/-пен*), and Genitive (*-ның/-нің/-дың/-дің/-тың/-тің*).
- **Verbal Negation and Aspect (12 allomorphs):** Root-adjacent unstressed negation infixes (*-ба/-бе/-па/-пе/-ма/-ме*, *-паған/-пеген*).
- **Evidentiality and Modality Markers (8 allomorphs):** Witnessed past (*-ды/-ді*), indirect evidential past (*-ыпты/-іпті*, *екен*), and necessity/possibility modal operators (*керек*, *тиіс*, *мүмкін*).

From the extracted morphological parses, the system constructs a continuous 16-dimensional morphological diagnostic vector $\mathbf{m}_{\text{affix}} \in [0,1]^{16}$ for any claim-evidence pair (c, E):

$$\mathbf{m}_{\text{affix}} = [m_1, m_2, \ldots, m_{16}]^\top \tag{2}$$

The vector specifically encodes: (1) Lexical and surface length ratios (m_1); (2) Directional negation mismatch $m_2 = \Delta_{\text{neg}} = |N_c - N_e|$; (3) Disjoint 4-digit calendar year conflicts $m_3 = \Delta_{\text{year}} = \mathbb{I}(\mathcal{Y}_c \cap \mathcal{Y}_e = \emptyset)$; (4) Quantitative numerical disparity $m_4 = \Delta_{\text{num}}$; (5–8) Evidentiality and epistemic modality mismatches (witnessed vs. reportative vs. speculative); (9) FST root Jaccard overlap $m_9 = J_{\text{root}} = \frac{|\mathcal{R}_c \cap \mathcal{R}_e|}{|\mathcal{R}_c \cup \mathcal{R}_e|}$; (10) Raw surface token overlap $m_{10} = J_{\text{surf}} = \frac{|\mathcal{S}_c \cap \mathcal{S}_e|}{|\mathcal{S}_c \cup \mathcal{S}_e|}$; (11) Inflectional divergence $\Delta_{\text{div}} = |J_{\text{surf}} - J_{\text{root}}|$; (12) Syntactic case root overlap $m_{12} = J_{\text{case}}$; and (13–16) Proper-noun coreference and sequence bounds.

4.2 Morpho-Contrastive AI Text Detection Module

To distinguish machine-generated Kazakh text from human-authored prose, we construct a Morpho-Contrastive detection network trained on a newly curated Tri-Domain dataset spanning 9,000 documents:

- **News Domain:** 3,000 articles from verified Kazakh journalistic agencies (Tengrinews, Kazinform, Azattyq).
- **Academic/Encyclopedic Domain:** 3,000 articles from the Kazakh Wikipedia repository.
- **Social/Conversational Domain:** 3,000 posts from Twitter, Telegram, and consumer discussion boards.

Machine-generated counterparts are synthesized using state-of-the-art LLMs, including GPT-4o, LLaMA-3-8B-Instruct, Mistral-7B, and Sherkala-7B (the leading open Kazakh foundation model).

The network encodes input text through a multilingual transformer backbone (XLM-RoBERTa-large) to produce contextual sequence embedding $\mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^d$. Simultaneously, the 83-rule FST computes the morpheme frequency distribution vector $\mathbf{v}_{\text{morph}} \in \mathbb{R}^K$. Both representations are projected into a shared metric embedding space $\mathbf{z} = \text{MLP}([\mathbf{h}_{[\text{CLS}]}; \mathbf{W}_m \mathbf{v}_{\text{morph}}])$. Training optimizes a joint objective combining Supervised Contrastive Loss ($\mathcal{L}_{\text{SupCon}}$) [28] and Cross-Entropy ($\mathcal{L}_{\text{CE}}$):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \, \mathcal{L}_{\text{SupCon}} \tag{3}$$

$$\mathcal{L}_{\text{SupCon}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \in A(i)} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)} \tag{4}$$

where $P(i)$ denotes positive instances sharing the same origin label (human vs. machine), and $\tau = 0.07$ is a temperature hyperparameter. The network outputs the generative authenticity score $T_{\text{gen}} \in [0,1]$:

$$T_{\text{gen}} = 1 - P(\text{AI-Generated} \mid c) \tag{5}$$

4.3 Morphologically-Grounded Fact Verification Pipeline

The factual verification tier operates across a curated knowledge corpus of 250,000 passage chunks extracted from Kazakh Wikipedia and public fact-checking archives (Factcheck.kz).

Hybrid Evidence Retrieval (Stage 1) To guarantee high evidence recall despite inflectional variance, retrieval fuses sparse lexical search and dense semantic representations:

(a) *Morpheme-Stemmed BM25*: Input queries are parsed through the 83-rule FST stemmer, mapping inflected tokens to canonical root lemmas: $c \to c_{\text{morph}}$. Standard BM25 scoring is computed over an inverted index of pre-stemmed passages:

$$S_{\text{BM25}}(c, e) = \sum_{t \in c_{\text{morph}}} \text{IDF}(t) \cdot \frac{f(t, e_{\text{morph}}) \cdot (k_1 + 1)}{f(t, e_{\text{morph}}) + k_1 \left(1 - b + b \cdot \frac{|e_{\text{morph}}|}{\text{avgdl}}\right)} \tag{6}$$

(b) *Dense Neural Retrieval (mContriever)*: Unsupervised multilingual dense representations compute bi-encoder cosine similarity:

$$S_{\text{dense}}(c, e) = \cos(\mathbf{h}_c, \mathbf{h}_e) = \frac{\text{Enc}(c) \cdot \text{Enc}(e)}{\|\text{Enc}(c)\| \|\text{Enc}(e)\|} \tag{7}$$

(c) *Reciprocal Rank Fusion (RRF)*: Candidate rankings are merged via RRF ($k_0 = 60$) [29]:

$$\text{RRF}(e) = \frac{1}{k_0 + \text{Rank}_{\text{BM25}}(e)} + \frac{1}{k_0 + \text{Rank}_{\text{dense}}(e)} \tag{8}$$

Cross-Encoder Verification (Stage 2) The top-k retrieved evidence sentences ($E = \{e_1, \ldots, e_k\}$) and the claim c are concatenated as a cross-encoder input sequence: $[c\,;\, E] \to \mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^d$. The contextual representation $\mathbf{h}_{[\text{CLS}]}$ is concatenated with the 16-D morphological diagnostic vector $\mathbf{m}_{\text{affix}}$ and projected through a linear classification head:

$$P(y \mid c, E) = \text{softmax}\left(\mathbf{W}_v \left[\mathbf{h}_{[\text{CLS}]}; \mathbf{m}_{\text{affix}}\right] + \mathbf{b}_v\right) \tag{9}$$

To counteract class imbalance, training optimizes weighted cross-entropy:

$$\mathcal{L}_{\text{NLI}} = -\sum_{k=1}^{3} w_k \cdot y_k \log P(y = k \mid c, E) \tag{10}$$

with inverse-frequency weights $w_{\text{SUP}} = 0.735$, $w_{\text{REF}} = 1.307$, $w_{\text{NEI}} = 0.958$.

4.4 Continuous 2D Cartesian Trust Matrix Formulation

To deliver actionable auditing intelligence, we formulate the continuous 2D Trust Matrix $\mathcal{T} = [-1, 1] \times [0, 1]$. We define the **Factual Veracity Score** $T_{\text{fact}} \in [-1, 1]$ as the calibrated difference between entailment and contradiction probabilities:

$$T_{\text{fact}} = P(\textsc{Supported} \mid c, E) - P(\textsc{Refuted} \mid c, E) \tag{11}$$

where $T_{\text{fact}} > 0$ denotes verified support, $T_{\text{fact}} < 0$ signifies refutation, and $T_{\text{fact}} \approx 0$ reflects insufficient evidence (NEI).

The **Generative Authenticity Score** $T_{\text{gen}} \in [0, 1]$ represents the probability of human authorship (Eq. 5). The unified scalar **Trust Metric** $\mathcal{T}(x) \in [0, 1]$ integrates both dimensions:

$$\mathcal{T}(x) = \sqrt{\frac{1}{2}\left(\max(0, T_{\text{fact}})^2 + T_{\text{gen}}^2\right)} \tag{12}$$

The continuous space induces four operational risk quadrants:

- **Quadrant 1 (Q1: Verified Human Ground Truth):** $T_{\text{fact}} > 0, T_{\text{gen}} \geq 0.5$ (Low Risk / Authoritative).
- **Quadrant 2 (Q2: Human Misinformation):** $T_{\text{fact}} \leq 0, T_{\text{gen}} \geq 0.5$ (Factual Error / Bias).
- **Quadrant 3 (Q3: AI Disinformation):** $T_{\text{fact}} \leq 0, T_{\text{gen}} < 0.5$ (Critical Threat / Synthetic Deception).
- **Quadrant 4 (Q4: AI Synthetic Knowledge):** $T_{\text{fact}} > 0, T_{\text{gen}} < 0.5$ (Synthesized Truth / Needs Attributed Audit).

5 Expected Outcomes and Major Innovations

5.1 Expected Tangible Deliverables

The execution of this research program is structured to produce high-impact theoretical and engineering deliverables, advancing the state-of-the-art in low-resource computational linguistics:

Del 1: Completed Master's Dissertation: A comprehensive 116-page academic dissertation formatted in strict compliance with Northwestern Polytechnical University graduate school standards, detailing theoretical foundations, architectural formulations, extensive empirical evaluations, and societal implications.

Del 2: The Kazakh-FEVER 3K Benchmark Dataset: The first human-validated, multi-domain fact verification dataset for Kazakh, comprising 3,000 balanced claim-evidence pairs across eight knowledge domains, annotated with sentence-level evidential citations and substantial inter-annotator agreement ($\kappa = 0.84$). The release includes a public 150-sample balanced benchmark split with standardized reproduction scripts.

Del 3: The 83-Rule FST Morphological Engine and Extractor: An open-source, deterministic finite-state transducer toolkit capable of sub-millisecond morphological segmentation, root lemmatization, and continuous 16-dimensional diagnostic vector extraction for Kazakh and adaptable to neighboring Kipchak Turkic languages.

Del 4: High-Level Peer-Reviewed Publications:

- A published conference paper at the *Artificial Intelligence and Natural Language Conference (AIST 2026)* documenting the morpho-contrastive AI text detection baseline and leave-one-generator-out evaluation.
- A full research paper submitted to the *ACL Rolling Review (ARR)* for commitment to the *31st International Conference on Computational Linguistics (COLING 2027)*, detailing the Kazakh-FEVER benchmark, the Hard NEI protocol, and the 2D Trust Matrix.

Del 5: Production-Grade Interactive Web Demonstration: A fully localized, bilingual (Kazakh and English) web interface deployed via Gradio and Hugging Face Spaces, featuring procedural SVG 2D Trust Matrix telemetry and real-time morphological stem highlighting for human-in-the-loop fact-checking.

5.2 Key Scientific Innovations

This research establishes three fundamental scientific innovations that address unresolved challenges in the NLP literature:

★ **Innovation 1: Deep Morpho-Structural Alignment for Agglutinative NLP.** Unlike conventional approaches that rely blindly on subword tokenizers or decoupled statistical stemmers, our framework

establishes a formal hybrid architecture. By directly fusing an 83-rule symbolic FST diagnostic representation ($\mathbf{m}_{\text{affix}} \in \mathbb{R}^{16}$) with dense contextual cross-encoder embeddings, our system explicitly preserves inflectional boundaries, resolves agent-patient syntactic role inversions, and neutralizes negation dilution.

★ **Innovation 2: The Hard NEI Curation Protocol for Shortcut Neutralization.** We identify and systematically eliminate the pervasive shortcut learning pathology in fact-checking benchmarks. By curating counterfactual claims that maintain over 85% lexical overlap with candidate passages while strictly manipulating modal, evidential, and temporal operators, the Hard NEI protocol forces models to learn truth-conditional reasoning rather than superficial word-matching heuristics.

★ **Innovation 3: Unified 2D Cartesian Trust Matrix Formulation.** We provide the first mathematical integration of factual veracity (T_{fact}) and generative provenance (T_{gen}) into a continuous Cartesian space. This innovation bridges two traditionally disjoint research fields, delivering an interpretable, operational risk taxonomy that empowers human editors to distinguish between human misinformation, AI synthesis, and adversarial AI disinformation.

6 Feasibility, Working Conditions, and Collaborative Environment

6.1 Computational and Infrastructure Feasibility

The computational demands of this research are fully supported by the high-performance computing facilities available at Northwestern Polytechnical University (NPU) and collaborative partner institutions. Model training, dense passage indexing, and large-scale experimental evaluations leverage dedicated GPU clusters equipped with NVIDIA A100 (80 GB VRAM) and Dual NVIDIA Tesla T4 (16 GB VRAM each) accelerators.

The project utilizes an open-source, highly reproducible software stack:

- **Deep Learning Frameworks:** PyTorch 2.3+ with CUDA 12.1, Hugging Face Transformers, Datasets, and Accelerate libraries, incorporating automatic mixed precision (`torch.amp.autocast`) and gradient checkpointing.
- **Information Retrieval Engines:** Apache Lucene-based Pyserini / Rank-BM25 for inverted index management, and FAISS for high-dimensional vector similarity indexing.
- **Morphological Transducers:** Standalone, dependency-light Python implementations of the 83-rule FST finite-state automaton, with C++ OpenFST bindings for high-throughput batch preprocessing.
- **Version Control and Reproducibility:** Git-managed codebases synchronized across private and double-blind anonymous academic repositories (e.g., Anonymous 4open science), with containerized deployment environments managed via Docker.

6.2 Data Resources and Linguistic Annotations

The empirical foundation of this study is grounded in extensive, publicly accessible knowledge repositories:

(1) **Kazakh Wikipedia Repository:** A filtered, orthographically standardized corpus of 250,000 text chunks extracted from official Wikimedia dumps (licensed under CC BY-SA 4.0), providing dense coverage across historical, scientific, geographic, and cultural domains.

(2) **Verified Public Fact-Checking Archives:** Real-world disputed claims and investigative fact-checks gathered from Factcheck.kz and regional news repositories, accessed under fair-use academic benchmarking standards.

(3) **Tri-Domain AI Detection Corpus:** 9,000 balanced articles across formal news, Wikipedia passages, and social media discussions, paired with synthetic generations from top-tier LLM families.

6.3 Academic Mentorship and Dual-Institutional Collaboration

This research operates at the intersection of deep learning and low-resource computational linguistics, benefiting from formal bilateral institutional collaboration:

◇ **Northwestern Polytechnical University (NPU), School of Computer Science:** Under the primary academic supervision of **Professor Bin Guo**, the candidate receives comprehensive mentorship in deep representation learning, ubiquitous intelligence, model generalization, and robust neural architectures. NPU provides the primary computational servers, academic guidance, and research infrastructure necessary for high-level dissertation execution.

◇ **Al-Farabi Kazakh National University (KazNU):** Under the co-advising and linguistic consultation of **Dr. Irina Ualiyeva**, the project coordinates with native Kazakh linguists and professional journalists. KazNU facilitates native speaker annotation campaigns, quality control cross-validation, and linguistic rule verification for the 83-rule FST analyzer.

This dual-institutional ecosystem provides an optimal balance between advanced machine learning methodologies and authoritative native linguistic grounding, ensuring the feasibility and rigorous completion of the proposed research.

7 Workload Estimation and Research Schedule

The execution of this Master's dissertation is structured across five sequential phases spanning the 2025–2027 academic cycle, ensuring the systematic completion of all theoretical, empirical, and writing milestones leading to formal degree defense in **June 2027**.

The total research workload involves approximately 2,200 hours of active investigation, encompassing data curation, code engineering, GPU training, empirical benchmarking, and thesis preparation, fully fulfilling the graduation credit requirements of Northwestern Polytechnical University.

8 Existing Problems and Proposed Countermeasures

In conducting this investigation, several technical challenges and methodological risks have been identified. Specific countermeasures have been designed to guarantee project success:

8.1 Challenge 1: Orthographic Divergence and the Ongoing Script Transition

Problem Description: The Kazakh language is currently undergoing a phased national transition from the Cyrillic script to a Latin-based alphabet. Web-scraped texts, social media posts, and official communications frequently exhibit mixed orthographic conventions, non-standard diacritics, and informal transliteration schemes. Standard tokenizers treat identical words in divergent scripts as completely disjoint vocabulary items.

Proposed Countermeasure: We introduce a bidirectional orthographic normalization pre-processing layer prior to FST parsing. The transducer maps informal Latin variants, Cyrillic homoglyphs, and archaic digraphs into a unified canonical Cyrillic-Kazakh character representation, ensuring invariant morphological segmentation regardless of input script.

Table 1: Dissertation Workload Allocation and Milestone Schedule (2025–2027)

Phase / Period	**Key Research Activities and Workload Focus**	**Deliverables / Milestones**
Phase 1 (Sept 2025 – Jan 2026)	Comprehensive literature review across AI text detection, FEVER paradigms, and Turkic morphology. Design and formalization of the 83-rule FST morphological analyzer. Ingestion and indexing of the 250,000-passage Kazakh knowledge corpus.	Formal literature survey; operational FST rule engine; indexed knowledge base.
Phase 2 (Feb 2026 – June 2026)	Execution of the Kazakh-FEVER 3K human annotation campaign under the Hard NEI protocol with inter-annotator validation. Implementation of the Morpho-Contrastive AI text detector on the Tri-Domain corpus. Preparation and publication of the AIST 2026 conference paper.	Kazakh-FEVER 3K benchmark; trained AI detection baseline; published AIST 2026 paper.
Phase 3 (July 2026 – Oct 2026)	Development of the hybrid BM25 + mContriever retrieval architecture and class-weighted cross-encoder verifier. Formulation of the continuous 2D Cartesian Trust Matrix $(T_{\text{fact}}, T_{\text{gen}})$. Submission of full research paper to the ACL Rolling Review (ARR) for COLING 2027.	COLING 2027 / ARR submission package; 2D Trust Matrix mathematical formulation.
Phase 4 (Nov 2026 – Feb 2027)	Integration of reviewer feedback from ARR cycle; execution of additional ablation studies and cross-generator robustness checks. Deployment of the bilingual Gradio interactive web demonstration. Completion of the initial full Master's dissertation draft.	Revision notes; live interactive web system; first complete dissertation draft.
Phase 5 (March 2027 – June 2027)	Dissertation formatting, internal supervisor review, and departmental pre-defense. External double-blind thesis review by the Graduate School academic committee. Final dissertation revision, archival submission, and formal Master's Degree Defense.	Dissertation pre-defense; final thesis approval; formal graduation in **June 2027**.

8.2 Challenge 2: Computational Latency of Dense Cross-Encoder Verification

Problem Description: While cross-encoders achieve superior accuracy by modeling bidirectional cross-attention across concatenated sequences ($[c\,;\,E]$), computing attention over thousands of candidate passages during open-domain verification incurs prohibitive computational latency, rendering real-time deployment infeasible.

Proposed Countermeasure: We implement a strict multi-tier retrieval filter. Stage 1 utilizes hybrid BM25 and quantized dense embeddings (mContriever) to prune the 250,000-passage knowledge corpus down to the top-k ($k = 5$) most salient candidate sentences within sub-second latency (< 300 ms). Full cross-encoder attention and 16-D morphological diagnostic feature extraction are restricted exclusively to this filtered candidate pool.

8.3 Challenge 3: Class Imbalance and Annotator Subjectivity on Hard NEI

Problem Description: In real-world news corpora, factual assertions naturally outnumber explicit refutations, causing severe class imbalance. Furthermore, classifying borderline Not Enough Info (NEI) instances involving subtle speculative or evidential markers introduces potential annotator subjectivity.

Proposed Countermeasure: To counteract class imbalance, cross-encoder fine-tuning optimizes a class-weighted cross-entropy loss function ($\mathcal{L}_{\text{NLI}}$) incorporating normalized inverse-frequency weights ($w_{\text{SUP}} = 0.735, w_{\text{REF}} = 1.307, w_{\text{NEI}} = 0.958$). To ensure annotation fidelity, all benchmark samples undergo dual independent cross-validation with formal inter-annotator agreement tracking ($\kappa = 0.84$), with conflicting labels resolved through adjudicated expert consensus.

9 Evaluation and Validation Methodology

The system will undergo rigorous quantitative and qualitative validation against recognized baseline architectures:

- **Retrieval Evaluation:** Evaluated across Recall@1, Recall@3, Recall@5, and Mean Reciprocal Rank (MRR) against standard BM25, morpheme-stemmed BM25, and dense MCONTRIEVER.
- **Verification Evaluation:** Evaluated across 3-class Accuracy, Macro-averaged F1, Hard NEI F1, and the official Strict FEVER score (which requires both accurate veracity prediction and gold evidence sentence retrieval) against fine-tuned mBERT-base, XLM-RoBERTa-base, XLM-RoBERTa-large, and zero-shot LLaMA-3-8B-Instruct.
- **AI Text Detection Evaluation:** Evaluated across Accuracy, F1-score, and Area Under the Receiver Operating Characteristic (AUROC) under strict Leave-One-Generator-Out (LOGO) cross-validation across diverse LLM families.

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备注

Reservation / Remarks:

The research program detailed in this title selection report reflects original, independent investigation conducted by the candidate under the direction of academic supervisor Prof. Bin Guo at Northwestern Polytechnical University, in academic consultation with Dr. Irina Ualiyeva at Al-Farabi Kazakh National University. All experimental protocols comply with ethical standards for open-source scientific research.